The infant was born in a frail condition at 43 weeks' gestation, approximately three weeks past the estimated due date. It remains unclear why labor had not been induced earlier. The infant's birth weight was 4 pounds, 3 ounces, placing it significantly below the expected weight for its gestational age. After the airway was cleared and residual amniotic fluid was removed, supplemental oxygen was administered immediately due to respiratory distress and poor oxygen saturation.

The mother's condition was also unstable following delivery. She presented with severe hypertension and substantial blood loss, requiring continuous observation and further intervention.

It is unclear how the infant's cranial abnormalities were not detected during prenatal ultrasound examinations. The structures may have been difficult to distinguish from developing bone, or their unusual appearance may not have been recognized as pathological by the ultrasound technician.

The infant was in a cephalic, head-first position when the mother was admitted. No major complications were identified during the initial assessment. However, while manually supporting the infant's head during delivery, I felt several hard, irregular structures beneath the skin surrounding the temporal region. Initially, I believed I had reached too far and was feeling the infant's shoulders. After confirming the position of my hand behind the infant's head, it became clear that the structures originated from the skull.

Following delivery, multiple hard, pale nodules were visible along both temples. They appeared to be embedded within the temporal bones rather than attached to the surface of the skin. The surrounding tissue was edematous, inflamed, and abnormally warm upon examination. The nodules were initially believed to be congenital bony protrusions. However, laboratory analysis subsequently identified their primary composition as synthetic polymers.

The structures are plastic deposits embedded directly within the infant's skull.

During observation in the neonatal intensive care unit, the skin surrounding several of the nodules ruptured. A pale, purulent discharge drained from the affected tissue, suggesting significant inflammation and possible secondary infection. The infant was started on broad-spectrum antibiotics and appears to be responding to treatment.

Post-Delivery Clinical Review

Radiographs and a computed tomography scan of the skull were performed once the infant's respiratory condition had stabilized. The results confirmed that the nodules are not superficial growths. Each is embedded directly within the temporal bones of the skull. The surrounding bone appears to have developed around the material during gestation, suggesting that the nodules were present long before delivery.

The largest nodule measures 2.4 centimeters across and penetrates approximately 7 millimeters into the skull. Several smaller deposits are distributed around it. They are pale, dense, and irregularly shaped, with smooth sections resembling manufactured plastic. Fine polymer filaments extend from the deposits into the surrounding bone. There is no clear boundary between the foreign material and the infant's natural tissue.

A biopsy was taken from an exposed portion of the right nodule. Laboratory analysis identified polyethylene, polypropylene, and several unknown synthetic compounds. These materials are commonly found in packaging, medical waste, and contaminated water.

There is no evidence that the nodules are moving or behaving independently. Their danger is structural. Both temporal bones are malformed, and pressure from the deposits has displaced nearby tissue. Surgical removal is not currently recommended. The plastic has become integrated with the bone, and attempting to extract it could fracture the skull or damage major blood vessels.

The fluid leaking from the split skin is consistent with inflammation and tissue breakdown around the exposed nodules. Cultures remain inconclusive, but antibiotics have been continued as a precaution. The wounds have been cleaned and covered with sterile dressings. Swelling has decreased slightly, although the skin remains hot and discolored.

The infant will remain in the neonatal intensive care unit for respiratory support and neurological observation. At present, the child responds to light, sound, and touch. No seizure activity has been recorded.

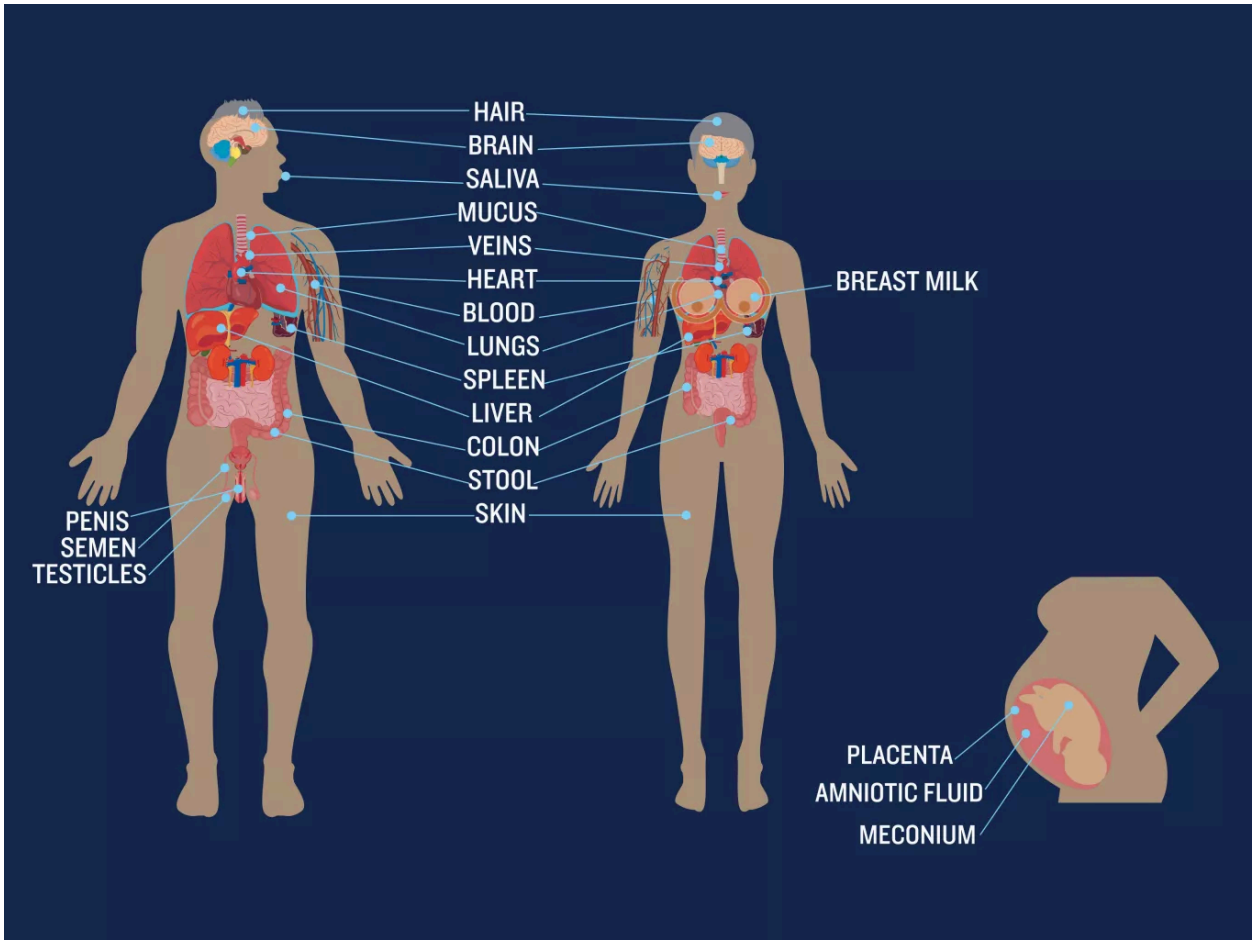
This is not an infection, a tumor, or a conventional congenital deformity. The infant's body did not simply reject the plastic.

It built the skull around it.



Children walking beside River Buriganga, in Dhaka, Bangladesh, where plastic waste floats in the water even after a recent cleanup

Credit: Syed Mahamudur Rahman/NurPhoto via AP



Microplastics in the human body

Credit: NRDC

Sources

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